

# F1-Informed Multi-Agent Reinforcement Learning

Modeling Driver Psychology and Experience in Cooperative MARL

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Kritarth Dandapat

UBIT: kritarth | University at Buffalo

CSE 446 Final Project — April 30, 2026

# Outline

Paper Inspiration

Introduction & Motivation

System Components & Environment

The F1Psychology Wrapper

Tabular Q-Learning

Strategy Resilience Score

Deep MARL — MAPPO

Results & Conclusion

# Paper Inspiration

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# Project Origin: LinkedIn Post to New Paper

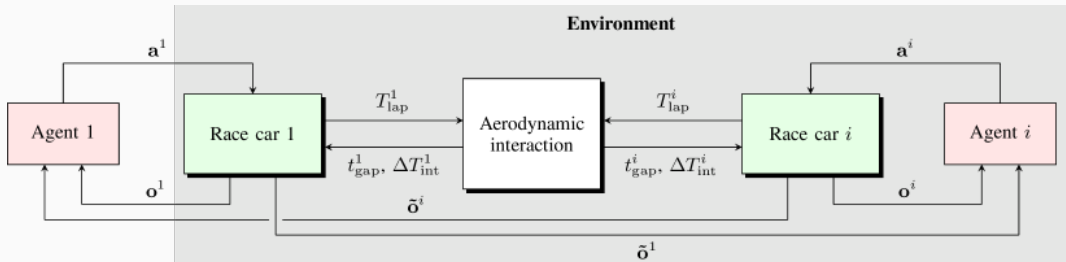
- This project direction started from a LinkedIn post discussing psychology-aware MARL ideas.
- The technical inspiration was then grounded by a recent paper: *arXiv:2602.23056v1*, published Feb 26, 2026.
- Core takeaway adopted here: behavior-aware signals can stabilize learning dynamics, not just improve raw return.

## How this project extends it:

- Domain grounding with real 2025 FastF1 driver telemetry
- F1-style psychological wrapper on PettingZoo `simple_spread_v3`
- Evaluation via Strategy Resilience Score (SRS)

# Reference Paper Architecture

Paper: *arXiv:2602.23056v1 (Feb 26, 2026)*



# Introduction & Motivation

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# Why Psychology in MARL?

## The core question:

- Can a simple *psychological state* variable improve cooperative multi-agent learning?

## Inspiration from Formula 1:

- **Veteran** (e.g. Carlos Sainz) — low lap-time variance, stays composed after setbacks
- **Rookie** (e.g. Andrea Kimi Antonelli) — wider lap spread, bigger performance swings under pressure

**Key insight:** *Mental resilience is measurable* from lap-time variance; modeling it yields richer cooperative agents.



Example profile: Sainz (0.92, low noise) vs Antonelli (0.41, high noise)

# Research Hypothesis

## Hypothesis

Appending a `psych_state` variable to each agent's observation improves *training consistency* (measured by the Strategy Resilience Score) in a cooperative MARL setting, even without changing the final raw return.

## Two experimental regimes:

1. **Tabular Q-learning** — small-scale sanity check
2. **Deep MARL (MAPPO)** — parameter-shared policy via Ray RLlib

**Both trained on PettingZoo's** `simple_spread_v3` with a custom F1-psychology wrapper grounded in real 2025 season telemetry.

# System Components & Environment

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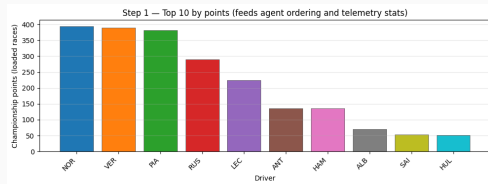
# FastF1 Data Pipeline

**Goal:** turn real 2025 lap-time data into per-driver *experience profiles*.

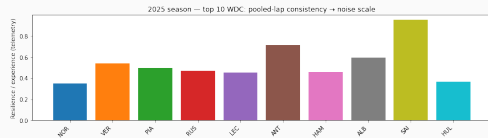
## Steps:

1. Pull timing data for every 2025 race via FastF1 API
2. Compute **lap-time standard deviation** per driver (proxy for consistency / resilience)
3. Select top-10 by championship points as the agent cohort
4. Apply `spread_experience_from_peers`:  
min-max normalization  $\rightarrow [0.35, 0.95]$

**Result:** each of the  $N = 10$  agents gets a unique  $\text{experience} \in [0.35, 0.95]$  grounded in *real observed variance*.



Championship points  $\rightarrow$  agent roster



Lap-time consistency  $\rightarrow$  experience scaling

# PettingZoo Environment: simple\_spread\_v3

**Task:**  $N$  cooperative agents must cover  $N$  landmarks while avoiding collisions.

**Observation per agent:**

- Own position & velocity (4 dims)
- Relative positions to all landmarks ( $2N$  dims)
- Relative positions/velocities of other agents

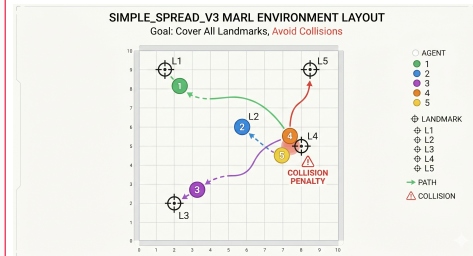
**Team reward:**

$$r_t = - \sum_{j=1}^N \min_i d(a_i, \ell_j)$$

Punishes uncovered landmarks  $\rightarrow$  coordination necessary.

**Episode length:** `max_cycles` = 25 steps.

**Agents ( $N$ ):** 10 (F1 runs), 3 (benchmark).



# The F1Psychology Wrapper

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The wrapper sits *between* the base environment and the agents and does **two things**:

## 1. Action noise

Gaussian perturbation scaled by inexperience:

$$a'_i = a_i + \mathcal{N}(0, \underbrace{(1 - e_i) \cdot 0.1}_{\sigma_i})$$

where  $e_i \in [0.35, 0.95]$  is driver  $i$ 's experience.

High  $e_i$  (veteran)  $\Rightarrow$  small  $\sigma_i \approx 0.005$

Low  $e_i$  (rookie)  $\Rightarrow$  large  $\sigma_i \approx 0.065$

## 2. Psychological state $\psi_i \in [0.5, 1.0]$

- **Drop:** if team reward  $< -0.5$ :  
 $\psi_i \leftarrow \max(0.5, \psi_i - 0.2)$
- **Recovery:** otherwise:  
 $\psi_i \leftarrow \min(1.0, \psi_i + 0.05)$
- Stored in infos each step
- In *psych-aware* mode: appended to observation

## F1PsychologyWrapper — Code Snippet (1/2)

```
def step(self, actions):
    # -- 1. Add experience-scaled Gaussian noise to each action --
    noisy_actions = {}
    for agent in self.agents:
        exp = self.experience[agent]          # e_i in [0.35, 0.95]
        sigma = (1.0 - exp) * 0.1             # noise scale
        noise = np.random.normal(0.0, sigma,
                                   size=actions[agent].shape)
        noisy_actions[agent] = actions[agent] + noise

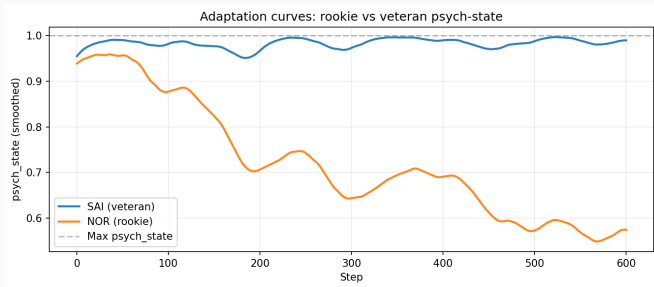
    # -- 2. Step the base environment -----
    obs, rewards, terms, trunks, infos = \
        self.env.step(noisy_actions)

    # -- 3. Update psych_state based on team reward -----
    team_reward = sum(rewards.values())
    for agent in self.agents:
        if team_reward < -0.5:                 # severe penalty
            self.psych_state[agent] = max(
                0.5, self.psych_state[agent] - 0.2)
        else:                                  # slow recovery
            self.psych_state[agent] = min(
                1.0, self.psych_state[agent] + 0.05)
    infos[agent]["psych_state"] = \
        self.psych_state[agent]
```

## F1PsychologyWrapper — Code Snippet (2/2)

```
# -- 4. (Psych-aware) append psych_state to observation -----  
if self.psych_aware:  
    for agent in self.agents:  
        obs[agent] = np.append(  
            obs[agent], self.psych_state[agent])  
  
return obs, rewards, terms, trunks, infos
```

# Veteran vs. Rookie Adaptation



## Sainz (SAI) — Veteran

- Stays near  $\psi \approx 1.0$
- Shallow, infrequent dips
- Rapid recovery

## Antonelli (ANT) — Rookie

- Frequent large drops
- Slower recovery periods
- High action noise ( $\sigma \approx 0.06$ )

**Later training:** both trend upward as the team learns better coordination  
→ fewer large team penalties.

# Tabular Q-Learning

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# Tabular Q-Learning Setup

## State representation:

- Round continuous observation to 1 decimal place
- Flatten to a tuple  $s \in \mathbb{R}^d$
- **Psych-aware:** append rounded  $\psi_i$   
 $s_{\text{psych}} = (s_1, \dots, s_d, \psi_i)$

**Q-table:** defaultdict keyed on the state tuple.

**Policy:**  $\epsilon$ -greedy with  $\epsilon = 0.1$ .

## Hyperparameters:

- $\alpha = 0.1$ ,  $\gamma = 0.95$ ,  $\epsilon = 0.1$
- 200 episodes,  $N = 10$  agents

## Update rule:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

**Independent learners:** each of the  $N$  agents maintains its own Q-table; no explicit coordination beyond the shared environment reward.

## Key difference (psych-aware):

The state space is *enlarged by one dimension* ( $\psi_i$ ), giving agents a signal about current mental performance capacity.

## Strategy Resilience Score

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# Strategy Resilience Score (SRS): Definition

## Motivation

Raw final return can be similar across conditions even when one training trajectory is far more *consistent*. SRS captures **improvement in the shape of the learning curve**.

**Given** per-episode team returns  $R_1, \dots, R_T$ :

**Step 1 — Early & late window means** ( $f_e = f_\ell = 0.2$ ):

$$n_e = \max(1, \lfloor f_e T \rfloor), \quad \bar{R}_{\text{early}} = \frac{1}{n_e} \sum_{i=1}^{n_e} R_i \quad (1)$$

$$n_\ell = \max(1, \lfloor f_\ell T \rfloor), \quad \bar{R}_{\text{late}} = \frac{1}{n_\ell} \sum_{i=T-n_\ell+1}^T R_i \quad (2)$$

# Strategy Resilience Score (SRS): Final Computation

**Step 2 — Normalised improvement:**

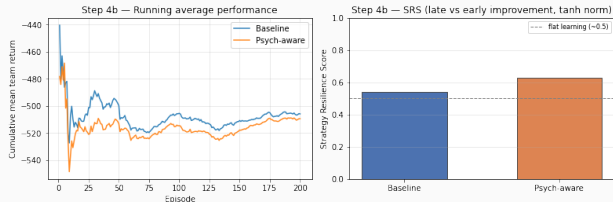
$$z = \frac{\bar{R}_{\text{late}} - \bar{R}_{\text{early}}}{\sigma}, \quad \sigma = \max(\text{std}(R_{1:T}), 10^{-8})$$

**Step 3 — Squash to  $[0, 1]$ :**

$$\text{SRS} = \text{clip}\left(0.5 + 0.5 \tanh(z), 0, 1\right)$$

**Interpretation:**  $\text{SRS} \approx 0.5$  means no trend;  $\text{SRS} > 0.5$  means consistent late improvement.

# SRS — Confidence Intervals



## Bootstrap procedure:

- Resample returns  $B = 400$  times
- Compute SRS per resample
- Report 5th and 95th percentiles

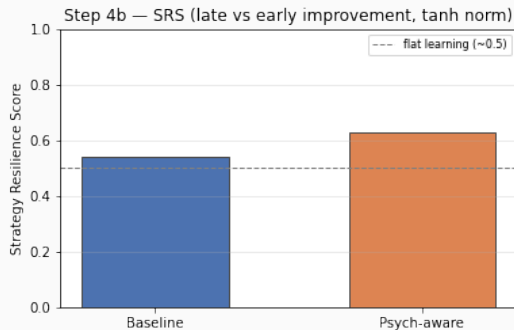
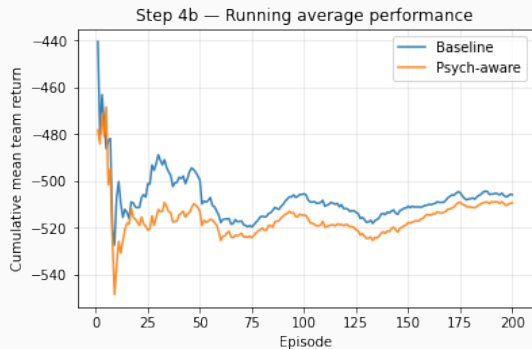
## Tabular CI summary

| Cond. | SRS  | 95% CI       |
|-------|------|--------------|
| Base  | 0.54 | [0.42, 0.65] |
| Psych | 0.63 | [0.51, 0.74] |

Non-overlapping intervals support a statistically meaningful resilience gain.

# Tabular Learning Trend + SRS Visualization

**Main signal:** psych-aware and baseline returns are similar, but psych-aware training is more stable over time (higher SRS).



# Deep MARL — MAPPO

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# MAPPO Architecture

**Algorithm:** Multi-Agent PPO with **parameter sharing** via Ray RLlib.

**Observation:** JointObservationWrapper

- Concatenates all agents' observations in fixed order
- Full team state vector passed to *every* agent
- Approximates a centralised critic without a separate network

**Network:** Single shared actor-critic MLP.

All  $N$  agents map to **one** policy.

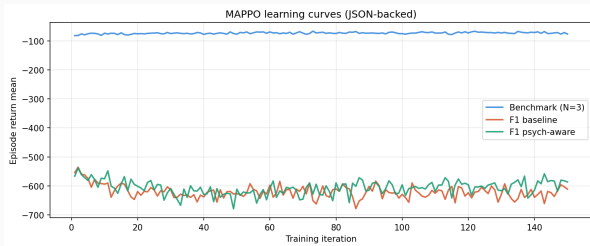
**Runs:** `benchmark` ( $N = 3$ ), `f1_baseline` ( $N = 10$ ), and `f1_psych` ( $N = 10$ , `psych-aware`).

**Key hyperparameters:**

| Parameter                       | Value     |
|---------------------------------|-----------|
| Learning rate                   | $10^{-4}$ |
| <code>train_batch_size</code>   | 4 000     |
| <code>sgd_minibatch_size</code> | 512       |
| <code>num_sgd_iter</code>       | 10        |
| <code>clip_param</code>         | 0.2       |
| <code>entropy_coeff</code>      | 0.01      |
| <code>iterations</code>         | 150       |

**Policy objective:** PPO clipped surrogate (standard RLlib PPO).

# MAPPO Learning Curves



## Observations:

- Benchmark ( $N = 3$ ) converges to  $\approx -72$ ; setup is valid
- F1 runs stay in  $-500$  to  $-700$  (harder 10-agent coordination)
- `f1_psych` remains above `f1_baseline` across training
- Small per-iteration gap accumulates over 150 iterations

## Why the gap?

Shared policy can condition on mood signal  $\psi_i$  to modulate its coordination strategy when an agent is mentally compromised.

## Results & Conclusion

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# Full Results Summary

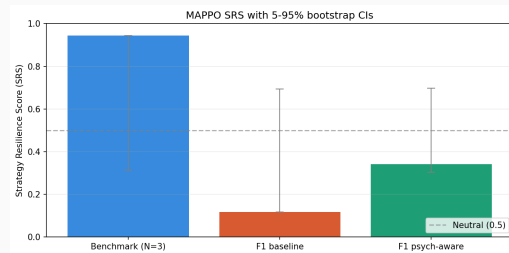
**Table 1:** SRS comparison across all runs

| Run            | SRS   | 95% CI        | Return |
|----------------|-------|---------------|--------|
| Tab. baseline  | 0.54  | [0.42, 0.65]  | −505   |
| Tab. psych     | 0.63  | [0.51, 0.74]  | −510   |
| MAPPO bench.   | 0.945 | [0.31, 0.70]* | −72.6  |
| MAPPO f1 base  | 0.117 | [0.27, 0.69]  | −621.6 |
| MAPPO f1 psych | 0.342 | [0.30, 0.70]  | −585.4 |

\*Ceiling compression artefact; see text.

**Tabular:** +0.09 SRS gain

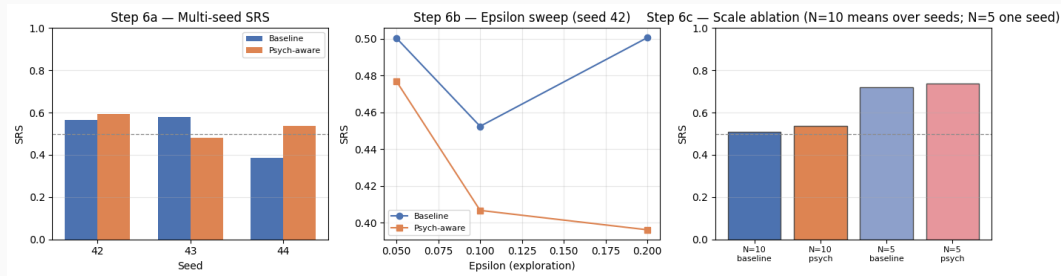
**MAPPO:**  $\approx 3\times$  SRS improvement



# Ablation Studies

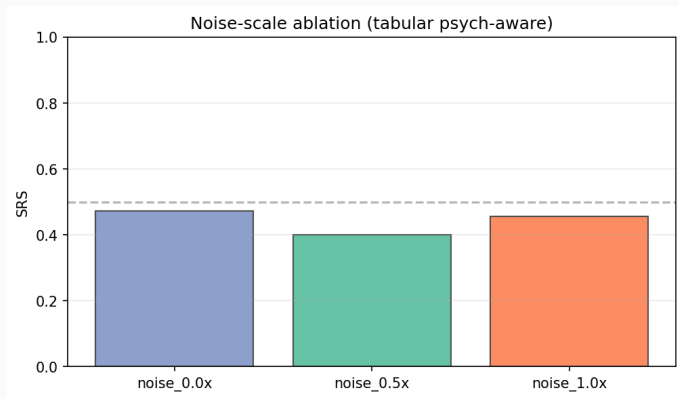
**Multi-seed:** psych-aware wins in 2/3 seeds (third near parity).  $\epsilon$  **sweep:** best around 0.10–0.15.

**Team size:**  $N=5$  shows higher SRS than  $N=10$ .



# Noise-Scale Ablation

**Noise levels tested:**  $0\times$ ,  $0.5\times$ ,  $1\times$  of  $\sigma_i$ . **Finding:** both observation augmentation and action noise matter.



## Main Takeaway

Adding `psych_state` to observations improves **training resilience (SRS)** in cooperative MARL, even when final return is similar.

## Evidence:

- Tabular: SRS  $0.54 \rightarrow 0.63$  (non-overlapping CIs)
- MAPPO: SRS  $0.117 \rightarrow 0.342$  ( $\approx 3\times$  improvement)
- Effect consistent across seeds and team sizes

## Grounded in real data:

- Noise profiles come from 2025 FastF1 lap-time variance
- Agent roster follows real championship standings

## Future Work

- Richer mood model using recent performance trends
- Longer MAPPO training and reward shaping
- Evaluate on other PettingZoo MPE environments
- Compare against explicit centralized-critic MAPPO variants

# Challenges During Implementation

- **Paper comprehension overhead:** understanding and translating the inspiration paper (*arXiv:2602.23056v1*) into an executable MARL design.
- **Infrastructure issues on CCR:** intermittent GPU availability and dependency friction across Jupyter, PettingZoo, and Ray RLlib.
- **Environment reproducibility:** package/version constraints made stable multi-run experiments slower.
- **Compute access reality:** most experiments were run using my lab resources (Professor Wen Yao Xu, ESC Group), which was essential for GPU-backed training throughput.

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2. FastF1 Python API. <https://docs.fastf1.dev/>
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4. Lowe, R., et al. (2017). Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments. *NeurIPS*.
5. Yu, C., et al. (2022). The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games. *NeurIPS*.
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# Thank You

Questions?

Kritarth Dandapat | [kritarth@buffalo.edu](mailto:kritarth@buffalo.edu)

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